

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1415

Woodward-Tap1 138 kV
May 29, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1415. The Interconnection Customer proposes a 404.4 MW wind facility interconnecting at the Woodward-Tap1 138 kV substation in the Virginia area. The request was received on March 03, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 490 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$69,013,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: RUCH – COMOX 138 kV line
Contingency: COMOX – RIVERVIEW 138 kV
Loading: 204.1% (Rating: 255 MVA)
DFAX: 100.0%
Pre-project loading: 45.4%
2. Facility: BEARSKIN – RUCH 138 kV line
Contingency: RIVERVIEW SUBSTATION – COMOX 138 kV
Loading: 203.3% (Rating: 253 MVA)
DFAX: 100.0%
Pre-project loading: 43.8%
3. Facility: WOODWARD-TAP1 – WATERFORD 138 kV
Contingency: APLINE – WOODWARD-TAP1 138 kV L
Loading: 184.4% (Rating: 215 MVA)
DFAX: 100.0%
Pre-project loading: 3.7%
4. Facility: APLIE – WOODWARD-TAP1 SUBSTATION 138 kV
Contingency: WATERFORD – WOODWARD-TAP1 138 kV Line
Loading: 180.7% (Rating: 219 MVA)
DFAX: 100.0%
Pre-project loading: 3.6%
5. Facility: KILLIAN – RIDGEWAY 138 kV
Contingency: APLINE – RIDGEWAY SUBSTATION 138 kV
Loading: 154.4% (Rating: 175 MVA)
DFAX: 100.0%
Pre-project loading: 77.0%

6. Facility: WATERFORD – WOODSMERE 138 kV L
Contingency: RIDGEWAY – APLINE 138 kV
Loading: 146.6% (Rating: 189 MVA)
DFAX: 100.0%
Pre-project loading: 67.2%
7. Facility: SOUTH CREEK – LAYFAIR 138 kV
Contingency: BEAR – CLOVER 345 kV
Loading: 120.7% (Rating: 238 MVA)
DFAX: 6.3%
Pre-project loading: 110.0%
8. Facility: WOODSMERE – FIELDALE 138 kV
Contingency: APLINE – RIDGEWAY 138 kV
Loading: 111.6% (Rating: 220 MVA)
DFAX: 100.0%
Pre-project loading: 72.5%
9. Facility: APLINE – RIDGEWAY SUBSTATION 138 kV
Contingency: WATERFORD – WOODSMERE 138 kV
Loading: 109.9% (Rating: 252 MVA)
DFAX: 100.0%
Pre-project loading: 50.4%
10. Facility: FIELDALE – IVAR 138 kV
Contingency: APLINE – RIDGEWAY 138 kV
Loading: 105.7% (Rating: 224 MVA)
DFAX: 100.0%
Pre-project loading: 74.8%
11. Facility: ORLANDO COGEN LP – PERTH 138 kV
Contingency: RIVERVIEW – COMOX 138 kV
Loading: 104.7% (Rating: 239 MVA)
DFAX: 24.6%
Pre-project loading: 63.0%
12. Facility: RIVER – PERTH SUBSTATION 138 kV
Contingency: COMOX – RIVERVIEW 138 kV Line
Loading: 103.5% (Rating: 236 MVA)
DFAX: 24.6%
Pre-project loading: 61.3%
13. Facility: LAYFAIR – RIV 138 kV
Contingency: RIVERVIEW – COMOX 138 kV
Loading: 102.7% (Rating: 232 MVA)
DFAX: 24.6%
Pre-project loading: 59.8%
14. Facility: DUNN SIDING – BIRCHWOOD 345 kV line
Contingency: BIRCHWOOD – LEAVITT SUBSTATION 345 kV
Loading: 100.6% (Rating: 1,312 MVA)
DFAX: 6.6%
Pre-project loading: 98.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
RUCH – COMOX 138 kV line	Rebuild line with higher-capacity conductor	\$13,576,000	100% (\$13,576,000)
BEARSKIN – RUCH 138 kV line	Rebuild line with higher-capacity conductor	\$15,543,000	100% (\$15,543,000)
WATERFORD – WOODWARD-TAP1 138 kV	Rebuild line with higher-capacity conductor	\$7,203,000	100% (\$7,203,000)
WOODWARD-TP1 – APLINE 138 kV	Rebuild line with higher-capacity conductor	\$3,660,000	100% (\$3,660,000)
KILLIAN – RIDGEWAY 138 kV	Rebuild line with higher-capacity conductor	\$4,504,000	100% (\$4,504,000)
WATERFORD – WOODSMERE 138 kV L	Rebuild line with higher-capacity conductor	\$2,757,000	100% (\$2,757,000)
SOUTH CREEK – LAYFAIR 138 kV	Rebuild line with higher-capacity conductor	\$22,818,000	22% (\$5,056,000)
WOODSMERE – FIELDALE 138 kV	Rebuild line with higher-capacity conductor	\$3,174,000	100% (\$3,174,000)
APLINE – RIDGEWAY 138 kV	Reconductor line	\$1,038,000	100% (\$1,038,000)
FIELDALE – IVAR 138 kV	Reconductor line	\$2,823,000	100% (\$2,823,000)
ORLANDO COGEN LP – PERTH 138 kV	Reconductor line	\$4,950,000	100% (\$4,950,000)
RIVER – PERTH 138 kV line	Reconductor line	\$2,099,000	100% (\$2,099,000)
LAYFAIR – RIV 138 kV L	Reconductor line	\$1,217,000	100% (\$1,217,000)
DUNN SIDING – BIRCHWOOD 345 kV line	Reconductor line	\$20,472,000	7% (\$1,413,000)
	Total allocated to project		\$69,013,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.